GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Question Bank 6.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.

1. Sides 5, 7, 8 and 15, 21, 24

Similar — every ratio is 3

15/5 = 21/7 = 24/8 = 3.

2. Sides 6, 9, 10 and 8, 12, 15

Not similar — 8/6 = 12/9, but 15/10 = 1.5

Two ratios are 4/3; the third is 3/2.

For questions 3 through 5, the two triangles are similar. Find the missing sides.

3. Sides 4, 9 and 11; the copy's 4-side is 10.

22.5 and 27.5

k = 2.5.

4. Sides 18, 24 and 30; the copy's 30-side is 5.

3 and 4

k = 5/30 = 1/6.

5. Sides 3, 7 and 8; the copy's 7-side is 28.

12 and 32

k = 4.

For questions 6 and 7, the triangles are similar.

6. A 6-ft man's shadow is 8 ft long. At the same moment a pole's shadow is 20 ft. How tall is the pole?

15 ft

k = 20/8 = 2.5; 6 × 2.5.

7. Ty's eyes are 4 ft up, 2 ft from a mirror on the ground. In it he sees the top of a tree 15 ft beyond the mirror. How tall is the tree?

30 ft

k = 15/2 = 7.5; 4 × 7.5.

Choose one.

8. Which triangle is similar to one with sides 4, 6 and 9?

- ○ A. 6, 8, 11
- ● **B. 8, 12, 18**
- ○ C. 8, 12, 16
- ○ D. 4, 6, 12

B

Every side × 2.

9. A 5-ft girl casts a 4-ft shadow. At the same moment a tree's shadow is 24 ft. How tall is the tree?

- ○ A. 19.2 ft
- ● **B. 30 ft**
- ○ C. 25 ft
- ○ D. 120 ft

B

$k = 24 \div 4 = 6$; 5×6.